

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.IT Semester V)
Data Visualization with Power BI and Tableau

TABLEAU DASHBOARD PROJECT

Video Games Sales Performance & Segmentation Dashboard

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Class / Batch	B.SC.IT
Subject	Data Visualization with Power BI and Tableau
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1. Project Title

Video Game Sales, Performance & Segmentation Analysis Dashboard using Tableau

2. Objective

The primary objective of this project is to design and develop an interactive Business Intelligence dashboard in Tableau to analyze historical video game sales. It converts multi-region retail transaction records into visual analytics to evaluate global revenue drivers, compare regional market sizes (North America, Europe, Japan, and Other markets), identify high-performing genres, measure the impact of critic feedback, and segment game titles into distinct clusters using machine learning clustering within Tableau.

3. Project Description

This project utilizes Tableau Desktop/Public to build a multi-dimensional analytics solution based on video game industry data. The dashboard features KPI cards, longitudinal trend lines, horizontal genre breakdowns, regional comparative rankings, top-grossing game summaries, and an unsupervised K -Means clustering model ($K=3$) comparing Critic Count/Critic Score against Global Sales. Interactive filters (such as Publisher) enable dynamic data slicing to understand market behavior across gaming platforms and demographics.

4. Dataset Details

Parameter	Details
Dataset Name	Video Games Sales as at 22 Dec 2016
Source	Kaggle
Number of Records	16719
Important Fields	Name, Platform, Year_of_Release, Genre, Publisher, NA_Sales, EU_Sales, JP_Sales, Other_Sales, Global_Sales, Critic_Score, Critic_Count

5. Tools & Technologies Used

- Tableau Desktop / Tableau Public (Worksheet visual design, calculated fields, data pivoting, and dashboard layout)
- Data Reshaping / Pivoting Engine (Transformed wide columns NA_Sales, EU_Sales, JP_Sales, Other_Sales into standardized normalized rows: Region and Regional Sales)
- Statistical Clustering (Tableau Analytics Pane: K -Means Clustering with 3 clusters)

6. Tableau Worksheets / Charts

Visualization	Purpose
KPI Cards	Total Sales, Total Profit and Total Quantity

Line Chart	Sales trend over time
Bar Chart	Sales by Category
Map / Bar Chart	Sales by Region or State
Top Products Chart	Top 10 products by sales
Interactive Filters	Year, Category, Region and other relevant fields

7. Calculated Fields

The following calculated fields can be used in the Tableau dashboard:

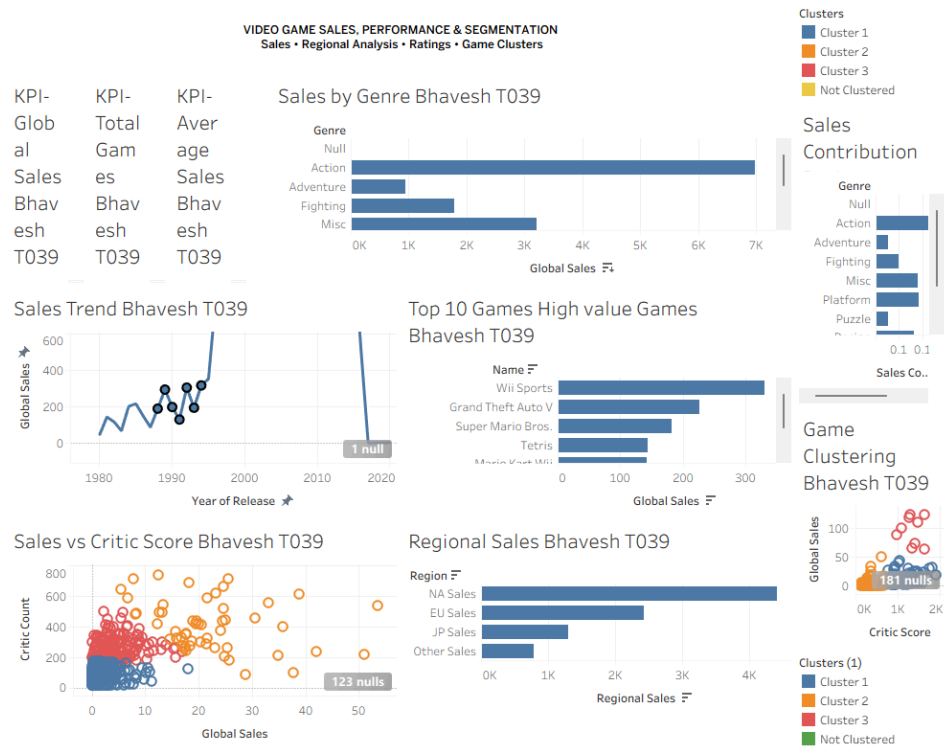
```
SUM([Global_Sales])

COUNTD([Name])

AVG([Global_Sales])

SUM([Global_Sales]) / TOTAL(SUM([Global_Sales]))
```

8. Dashboard Screenshot



Paste the final dashboard screenshot in this section.

9. Key Insights

1. North American Market Dominance:

The North American market (NA Sales) constitutes 4,400M units, which accounts for approximately 49.3% of all global sales. Europe follows as the second-largest territory with 2,420M units (27.1%), while Japan represents 1,290M units (14.5%). Commercial game publishing must prioritize NA and EU rating and localization pipelines first to capture over 76% of total returns.

2.Genre Concentration in Action, Sports, and Shooters:

Action is the top-grossing category at ~1,750M units sold (~19.5% overall market share), followed by Sports (~1,330M units, 15.0%) and Shooter (~1,050M units, 11.8%). These three genres together drive nearly half of the entire video game industry's revenue. Specialized categories such as Strategy (175M units) and Puzzle (245M units) have limited mainstream commercial volume.

3. Golden Era Boom and Seventh-Generation Hardware Peak (2006–2010):

The longitudinal sales line chart reveals exponential market expansion starting in 1996, climbing to an all-time industry peak between 2007 and 2009 (exceeding 670M units annually). This corresponds directly to the widespread adoption of the Nintendo Wii, PlayStation 3, and Xbox 360 generations.

4.Pareto Outlier Best-Sellers (Top 10 Concentration):

Commercial returns follow an extreme right-skewed distribution. While the mean game sale is only 0.5335M copies, the top 10 titles generate massive outlier revenues. Wii Sports leads the historic ranking (over 330M aggregated platform units), followed by Grand Theft Auto V and Super Mario Bros.

5.Actionable Segmentation via \$K\$-Means Clustering:Using Tableau's native machine learning clustering on SUM(Critic Score) and SUM(Global Sales) successfully segmented the catalog into three distinct commercial tiers:Cluster 1 (Blue): High-volume, low-margin catalog games with moderate review coverage and sales $\$ < 20 \times 10^6$.Cluster 2 (Orange): Mid-tier commercially stable performers generating consistent returns.Cluster 3 (Red): Elite, high-critic-score flagship blockbusters that achieve high commercial velocity.

10. Video Demonstration

Video Link: <https://youtu.be/NJ6MgZoAeSE>

11. GitHub Repository

GitHub Link:

<https://github.com/Frizzy1826/Data-Visualization-with-Power-BI-and-Tableau/tree/main/project>